



Lab 1: Modelling of the Motor System

MTE 484: Digital Control Systems

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1 Introduction

This lab introduces the motor and gear mechanism which are to be used for the 'ball balancing on beam' project, which must be properly characterized in order to facilitate the completion of the project. This includes experimentally determining values for 'stiction' and the transfer function of the motor. These activities were performed during the span of the lab session, and their procedures and results are described in detail in the following report.

2 Question C

By this point in the lab, the set up was already complete. The Arduino was functioning as intended, and with the sample code provided, it was possible to both send voltage values to the motor and read sensor data, which included the motor rotational position and the ball's position on the beam. These values were printed out on the serial monitor, allowing them to be recorded.

The values of the motor position being output needed to be converted to radians in order to be useful. Using the following formula, the motor position output values could be linearized and converted into radians:

$$p_{radians} = m * p_{sensor} + b \quad (1)$$

Where the m and b values must be determined through a system of equations, using experimental values of p_{rad} and p_{sensor} to do so.

The position of the motor was manually changed to three points: 0, $\frac{\pi}{4}$, and $-\frac{\pi}{4}$ radians. The values printed into the serial monitor were read at each point, providing concrete values of p_{rad} and p_{sensor} . These values were put into the system of equations, and solving for this provided values for m and b :

$$m = -0.000359453251, b = 1.75789534 \quad (2)$$

3 Question D

The term 'stiction' describes the concept of static friction that exists in motors before they start moving. This creates a range of voltage values that, when sent to the motor, will not move the motor at all. This impacts the accuracy of the mechanism, especially when considering the fine movements required to balance a ball on a beam. This must be counteracted by creating a 'deadzone' at a certain range; for example, if the deadzone range is determined to be -0.1 to 0.1 volts, any voltage input within this range would be immediately rounded to either 0.1 or -0.1, depending on the direction.

The values of voltage that determine the deadzone range must be experimentally determined. Additionally, it is important to determine them in both directions, as the stiction in motors is not often symmetrical. In order to complete this task, a function was derived that, given a direction, methodically applies an increasing voltage to the motor, and only stops the loop once the potentiometer detects the motor start to change position.

The values for stiction have been determined for both directions, differentiated by whether the voltage values were increasing or decreasing:

$$v_{increasing} = 0.120V, v_{decreasing} = 0.101V \quad (3)$$